CHAPTER TWO

Water use efficiency in mangroves: Conservation of water use efficiency determined by stomatal behavior across leaves, plants, and forests

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Abstract

Absorption of water from seawater with its high osmotic pressure is costly to mangroves and requires the plants to use water conservatively. Indeed, measurements of gas exchange, growth rate and carbon isotope composition have shown that mangrove species operate with higher water-use efficiency (WUE) than do most C_3 species. These characteristics of water use and water use efficiency find expression in interspecific differences in salinity tolerance, and in the structure of mangrove forests along complex gradients in salinity and aridity. Under global warming, mangroves face increasing challenges of varying salinity, humidity and temperature, all factors that affect water use and WUE. The extent to which mangroves can acclimate to changing environmental conditions by altering water-use characteristics depends on spatial (leaf, plant, and ecosystem) and temporal (short- and long-term) factors. Here, we clarify concepts of WUE at each spatiotemporal scale, compare their values across scales,

Advances in Botanical Research, Volume 103
ISSN 0065-2296
https://doi.org/10.1016/bs.abr.2022.02.017

and show consistency in maintenance of a high water use efficiency from leaves to ecosystems. We develop a framework for comparison of WUE across scales and identify areas for future research.

1. Introduction

Mangrove forests are most luxuriant and diverse along the tidal, saline coastal wetlands of the humid tropics and subtropics (Tomlinson, 2016). As halophytes, mangroves characteristically are conservative in water use (Ball & Passioura, 1995), operating with high carbon gain with respect to water expenditure. These water-use characteristics become increasingly conservative with increase in salinity and in the salt tolerance of the species (Ball, Cochrane, & Rawson, 1997; Ball, Cowan, & Farquhar, 1988). This may be a consequence of both high carbon costs of water uptake by the roots and restrictions in the rates of water usage by the leaves to maintain favorable water and salt relations (Ball, 1988; Ball et al., 1988). Maximizing carbon gain relative to water use is a whole-plant phenomenon that integrates stomatal behavior in relation to photosynthesis, physical properties and display of leaves in relation to light interception and evaporative demand, and the partitioning of carbon between structures supplying and those consuming carbon-based assimilates (Cowan, 1982; Cowan & Farquhar, 1977). Water-use characteristics thus find expression at all levels of plant form and function, ultimately contributing to variation in mangrove forest structure along natural gradients in salinity and aridity (Ball, 1988, 1996). As the improvement in leaf-level WUE may not be always translated into plant- or ecosystem-level WUE (Medrano et al., 2015), water-use efficiency for each scale should be included in comprehensive comparison of WUE among species. The present review builds on this knowledge to develop a framework enabling linkage of mangrove water use characteristics across scales from leaves to ecosystems.

The trade-off between carbon gain and water loss, water use efficiency (WUE) has been defined as the ratio of the amount of carbon assimilated and the amount of water transpired,

$$\mathrm{WUE} = \frac{\mathrm{CO_2\ fixed}}{\mathrm{H_2O\ transpired}} \tag{1}$$

Several kinds of WUE have been used to compare water use strategies of mangrove species, but there are limitations for application of each WUE.

Here, we first clarify concepts underlying the WUE terms usually used at specific spatiotemporal levels and then examine consistency in estimations of WUE between different spatiotemporal levels.

2. The definitions of water use efficiency

Plant leaves obtain CO_2 through open stomata but inevitably lose H_2O. The gas fluxes can be characterized by a one-dimensional form of Fick's first law of diffusion, i.e., conductance × concentration gradient. The assimilation rate (A) of carbon is thus $A = \frac{g}{1.6}(p_a - p_i)/P$, and transpiration rate (T) is $T = g(e_i - e_a)/P$, where P is air pressure, and $p_a - p_i$ is the drawdown of the intercellular and atmospheric CO_2 pressure, and $e_i - e_a$ is the drawdown of the intercellular and atmospheric vapor pressure. Indeed, WUE has been used interchangeably across spatiotemporal scales to refer to observations ranging from gas exchange by individual leaves for a few minutes to growth responses to precipitation through an entire season or lifecycle of the plant. However, WUE is a deceptively simple concept that requires clarification for use at different spatial and temporal scales (Table 1).

Consider, for example, that WUE calculated from leaf gas exchange does not capture leaf level adjustments in water use and water use efficiency that accompany acclimation of leaf photosynthetic capacity to short-term change in environmental conditions, nor does it capture the integrated carbon gain and water loss over longer time scales significant to growth (Medrano et al., 2015). An example is shown in Fig. 1, in which WUE in leaves of *Avicennia marina* calculated from instantaneous gas exchange data increased with gradual variation in salinity from 50 to 500 mM NaCl and then declined upon return to 50 mM NaCl (Fig. 1A). In contrast, WUEs calculated from biomass accumulation and total plant water loss during long term growth were similar under constant salinities of 50 and 250 mM NaCl and increased slightly in plants grown under 500 mM NaCl (Fig. 1B). In this example, acclimation to long term growth conditions reduced the variation in WUE between salinity treatments relative to the short-term exposure to changing salinities.

Differences in computation of WUE across scales are summarized in Table 1. These differences are particularly important when comparing values of WUE between species and environments, over short- and long-term time scales significant to daily integrated carbon gain of a leaf or to plant growth, and finally to integrated carbon/water balances relevant to ecosystem function.

Table 1 Definitions of water use efficiency (WUE).

Level	Long-term WUE	Instantaneous WUE	Intrinsic WUE
Leaf[1]	$\mathrm{WUE}_l = \frac{(1-\Phi)\int_{t1}^{t2} As\ \mathrm{d}t}{\int_{t1}^{t2} Ts\ \mathrm{d}t}$	$\mathrm{WUE}_l^{in} = \frac{A}{T} = \frac{p_a - p_i}{1.6(e_i - e_a)} \cong \frac{p_a - p_i}{1.6v}$	$\mathrm{IWUE}_l = \frac{A}{g} = \frac{C_a - C_i}{1.6}$
Plant[1]	$\mathrm{WUE}_p = \frac{(1-\Phi)\int_{t1}^{t2} A's'\ \mathrm{d}t}{\int_{t1}^{t2} T's'\ \mathrm{d}t}$	$\mathrm{WUE}_p^{in} = \frac{A'}{T'} \cong \frac{p_a' - p_i'}{1.6v}$	$\mathrm{IWUE}_p = \frac{A'}{g'}$
Ecosystem[2]	$\mathrm{WUE}_e = \frac{\int_{t1}^{t2} NPPs''\ \mathrm{d}t}{\int_{t1}^{t2} T''s''\ \mathrm{d}t}$	$\mathrm{WUE}_e^{in} = \frac{GPP}{T''}$	$\mathrm{IWUE}_e = \frac{GPP}{g''}$

Notes: A and T are assimilation rate and transpiration rate. The ' and '' refer to plant-level and ecosystem-level counterparts, and the subscripts l, p and e indicate the leaf-, plant-, and ecosystem-level WUE. e_i and e_a are the intercellular and atmospheric vapor pressure, and p_i and p_a are intercellular and atmospheric CO_2 pressure, $(p_a - p_i)/P = C_a - C_i$, and P is air pressure and s is leaf area. The g represents the total of boundary layer and stomatal conductance (1/resistance) to vapor, but the boundary layer resistance is much smaller than stomatal resistance especially for measurements using conventional gas exchange techniques. So g here indicates stomatal conductance for convenience, and the factor 1.6 is the ratio of diffusivities of water vapor and CO_2 in air. The Φ indicates the proportion of carbon that is fixed during the day but respired by the leaf at night and by other parts of the plant over the whole period. The net primary productivity (*NPP*) is the difference of gross primary productivity (*GPP*) and autotrophic respiration (*Ra*). The WUE_p can be estimated by the ratio of mass growth rate/water loss rate. The superscripts 1 and 2 indicate, respectively, Farquhar and Richards (1984) and Beer et al. (2009).

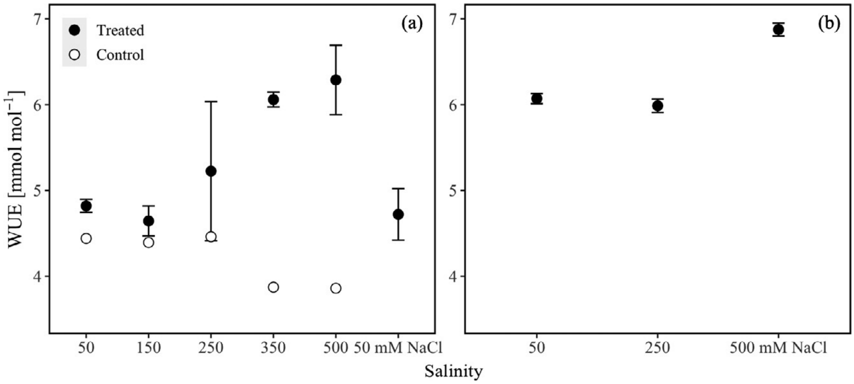

Fig. 1 Short-term (A) and long-term (B) responses of water-use efficiency (WUE) in *Avicennia marina* to variation in salinity under a constant humidity of 70% (12 mbar vpd). In (A) seedlings were grown hydroponically in nutrient solution amended with 50 mM NaCl (roughly 10% seawater salinity). Gas exchange measurements were made concurrently on fully expanded leaves under the growth conditions, and then measured repeatedly as the salinity was either gradually increased to 500 mM NaCl and then decreased gradually to the original conditions (solid circles, mean ± SE, $n = 3$ seedlings) or kept at 50 mM NaCl (hollow circle, n = 1 seedling). The short-term WUE was calculated from Ball and Farquhar (1984a) as $(1 - 0.35) \times A/T$ where 0.35 is an assumed proportion of dark respiration (Φ in Table 1) to facilitate comparison with the long-term results shown in panel (B). Here, seedlings (n = 5 per salinity) were grown hydroponically for three months in nutrient solution amended with 50, 250 and 500 mM NaCl. The long-term WUE were estimated from data in Ball and Farquhar (1984b).

3. Long-term vs. short-term water use efficiency

The actual WUE is an index describing the long-term (over the course of a whole day) carbon gain per unit water loss of plants, i.e., the spatial and temporal integration of A divided by that of T (Farquhar, O'Leary, & Berry, 1982). However, the instantaneous WUE (WUE^{in}) is commonly adopted instead of the long-term WUE for two reasons: first, measurements of components of the WUE^{in} are easier to obtain than those of the long term because of the continuous monitoring required for the latter; second, the instantaneous WUEs are identical to long-term ones if the drawdown of both CO_2 ($p_a - p_i$) and water vapor ($e_i - e_a$) through a leaf stay constant during daytime (steady state) and night-time autotrophic respiration is negligible ($\Phi = 0$ in Table 1). The rare availability of steady state conditions under naturally varying field conditions (radiation, temperature and humidity) hinders direct estimations of the long-term WUE, while a short-term controlled steady state is easy to reach with a portable photosynthesis system. In this context, we can consider the WUE^{in} as an average state and then add the factor $(1-\Phi)$ to the WUEin, i.e., $(1-\Phi)$WUEin if night-term autotrophic respiration is significant (Farquhar, O'Leary, & Berry, 1982). This greatly extends applications of instantaneous WUEs as approximate estimations of long-term WUEs. In addition, eddy covariance techniques facilitate long-term continuous measurements of carbon and water fluxes at the ecosystem level, but challenges exist in calculating long-term WUEs due to difficulties in distinguishing canopy transpiration flux ($T^{''}$ in Table 1) from evapotranspiration. However, $T^{''}$ can be estimated by sap flow techniques (Krauss, Barr, Engel, Fuentes, & Wang, 2015) and some process-based ecological models (Liang et al., 2019; Wei, Lee, Wen, & Xiao, 2018).

Aside from the models, there are two other commonly used approaches to evaluating long-term WUEs: one is destructive measurements of plant growth and water loss (ratio of mass growth rate/water loss rate), which is usually done with potted plants growing in controlled environments (Ball, 1988); the other is obtained by measurement of carbon isotope composition (^{13}C) of plant organic matter, as clarified in experimental and theoretical studies (Farquhar et al., 1993; Farquhar, Ball, Caemmerer, & Roksandic, 1982; Farquhar, O'Leary, & Berry, 1982; Farquhar & Richards, 1984). Importantly, leaf carbon isotope composition reflects the carbon component of water use efficiency integrated over the past environment of the leaf. It therefore is a long-term concept, largely decoupled from the instantaneous WUE estimated by simultaneous gas exchange measurements (Martin et al., 2010; Medlyn et al., 2017). Farquhar, O'Leary, and Berry (1982) derived one theory relating WUE to carbon isotope composition of leaf dry matter for C_3 plants as

$$WUE = \frac{(1-\Phi)C_a}{1.6\bar{D}} \frac{b-\delta_a+\delta}{b-a} \tag{2}$$

where the discrimination, a, associated with diffusion in air is 4.4‰ (Farquhar, O'Leary, & Berry, 1982). The discrimination in the carboxylation reaction, b, is 27‰ as determined from comprehensive statistics of many plants, including mangroves, with the assumption that the intercellular CO_2 mol fraction C_i is the same as that (C_c) at carboxylation sites (Farquhar, Ball, et al., 1982). More complex calculations have been developed, see Busch, Holloway-Phillips, Stuart-Williams and Farquhar, 2020. The δ_a is carbon isotope composition of atmospheric CO_2 relative to the VPVB standard (−8.7‰). The $\overline{D}$ is a weighted mean vapor mole fraction difference (v/P). In general, the greater carbon isotope ratio in mangrove leaves indicates higher water use efficiency (Farquhar, Ball, et al., 1982; Liang et al., 2018; Lin & Sternberg, 1992) than in neighboring plants growing under the same atmospheric conditions (temperature and humidity). If the plants compared grow under different environmental conditions, then Eq. (2) should be adapted with inputs of specific environment factors to estimate WUE. In addition, without Φ included in Eq. (2), WUE will be overestimated relative to measurements of the leaf-level instantaneous WUE (Medlyn et al., 2017).

4. Instantaneous WUE vs. intrinsic WUE

The ratio of assimilation rate over stomatal conductance, called the intrinsic WUE (IWUE), is only related to CO_2 drawdown while the instantaneous water use efficiency is related to the drawdown of both CO_2 and water vapor. This intrinsic term was first introduced to compare photosynthetic properties at a common evaporative demand (Osmond, Björkman, & Anderson, 1980). Since then, IWUE has been used in inter- and intra-species comparisons (Niinemets, Diaz-Espejo, Flexas, Galmes, & Warren, 2009), as it is thought the IWUE (Table 1) is only determined by intrinsic photosynthetic (biochemical) characteristics of vascular plants (Niinemets, Flexas, & Penuelas, 2011), such as mesophyll conductance and Rubisco properties (Flexas et al., 2016). However, IWUE and its component C_i vary with D. Consider the example shown in Fig. 2 for whole plants of *Avicennia marina* grown under different salinity x humidity conditions. The linear

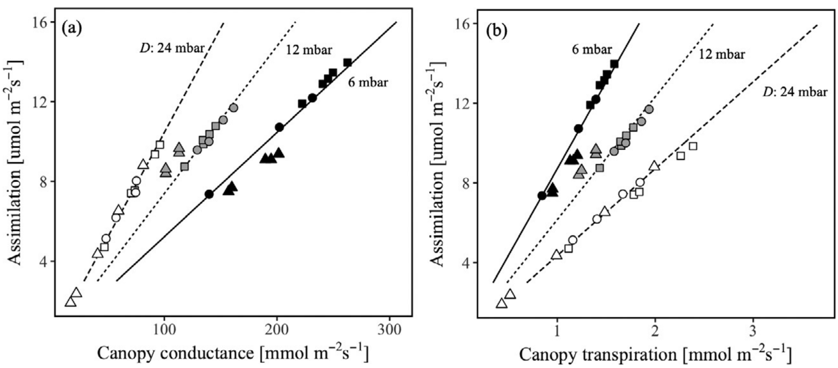

Fig. 2 Assimilation rate as a function of plant level canopy conductance (A) and canopy evaporation rate (B) in seedlings of *Avicennia marina* grown hydroponically in nutrient solution amended with 50 (diamond), 250 (circle) and 500 mM NaCl (triangle) under high, medium or low humidity where D was 6, 12 or 24 mbar, respectively. Each point within a panel represents a different plant with the same data set used for both panels. The assimilation rates were here estimated from Ball (1988) as mass growth rate $\times$ 0.45/12 assuming that the carbon content of the whole mangrove seedling is 0.45% of dry matter (Ma et al., 2018). Lines indicate best fits.

relationships between A and g for plants grown in a range of salinities under a particular D, indicate roughly constant C_i, and hence also WUE (Fig. 2A). These results showed that D had a greater effect on WUE than did salinity. Now compare the patterns in IWUE revealed by A/g (Fig. 2A) with those of actual WUE revealed by plots of A as a function of transpiration (Fig. 2B). Note that the IWUE (Fig. 2A) was greatest in plants grown under low humidity (24 mbar D), while the actual WUE (Fig. 2B) was lowest. This occurred because reduction in stomatal conductance with increasing D was not sufficient to reduce the transpiration rate. It increased with increasing D, although T under high D was not as great as it would have been if stomata had not partially closed. Nevertheless, the actual WUE declined with increasing D. This comparison underscores the importance of including estimates of water use in calculations of WUE. Specifically, comparisons of IWUE values are valid if measurements are made under common D, leaf temperature and ambient CO_2 concentration. A lack of awareness of this limitation has influenced some popular perceptions in the literature. For example, Clough and Sim (1989) measured A/g in mangroves growing naturally in forests along an aridity gradient and concluded that increase in A/g showed that WUE increased under environmental stress due to aridity. If they had plotted A/T, then they would likely have concluded the opposite.

In contrast to IWUE, which varies with D, the marginal water cost, $\lambda = \partial T/\partial A$ is constant with diurnal variations in air temperature and D over the course of a day (Cowan, 1982; Cowan & Farquhar, 1977; Farquhar, Schulze, & Kuppers, 1980). Consequently, λ is a more appropriate index for comparisons of water-use efficiency between different species or the same species growing under different atmospheric conditions. A higher λ implies greater benefit to the plant of carbon gain or lower cost to the plant of water supply to leaves. Short-term maintenance of λ can be achieved through adjustments in stomatal conductance under varying atmospheric conditions, but longer term adjustments of λ also occur with acclimation of photosynthetic or hydraulic characteristics to changing environmental conditions (Cowan, 1978; Cowan & Farquhar, 1977). Challenges remain in distinguishing short-term stomatal behavior from long-term acclimation.

5. Compromise between WUE and photosynthetic nitrogen use efficiency

In their seminal paper, Field and Mooney (1986) showed that a high WUE comes at the expense of increasing the N cost of carbon gain (i.e., PNUE), especially in C_3 species, which includes all mangroves.

Decreasing stomatal conductance reduces water loss more than assimilation rates, thereby increasing WUE for a given photosynthetic capacity, but at the expense of carbon gain. In contrast, increasing photosynthetic capacity increases both the assimilation rate and WUE for a given stomatal conductance. Enhancing photosynthetic capacity does not compromise carbon gain, but high WUE is achieved at the expense of nitrogen-use efficiency (NUE), because more nitrogen (N) must be invested in photosynthetic machinery, notably ribulose 1·5-bisphosphate carboxylase/oxygenase (Rubisco) (Evans, 1989). Hence, it follows from Field and Mooney (1986) that increasing WUE in response to increasing salinity would either compromise carbon gain or increase the N cost of carbon gain.

The trade-off between WUE and PNUE (Field & Mooney, 1986) can explain gas exchange results of field-based studies of effects of N-fertilization on individual mangrove trees growing naturally in N-limited systems. Leaf level WUE was greater in fertilized than control trees (Lovelock, Feller, McKee, Engelbrecht, & Ball, 2004; Martin et al., 2010) and increased with increasing soil water salinity (Martin et al., 2010). The N-fertilized leaves had greater N contents than controls, but there was no significant relationship between leaf N contents and photosynthetic assimilation rates (Lovelock et al., 2004; Lovelock & Feller, 2003; Martin et al., 2010). The N-fertilized leaves must have invested greater amounts of N in photosynthetic machinery to account for the maintenance of assimilation rates with reduction in stomatal conductance. However, such investment may also have been limited at high salinity by competing demands for N to meet other physiological needs. For example, osmotic adjustment of cytoplasmic compartments and other protective functions may require greater synthesis of N-containing compatible solutes with increasing salinity (Popp, 1984; Popp & Polania, 1989). Nevertheless, the greater investment of N in photosynthetic machinery by N-fertilized trees subsidized more conservative use of water without penalizing carbon gain per unit leaf area, thereby enabling limited water resources to support photosynthetic carbon gain over a larger canopy area (Martin et al., 2010).

6. Into the future: Response of water-use efficiency to elevated CO_2

Elevated atmospheric [CO_2] can enhance WUE, if a higher or similar assimilation rate can be sustained at a lower stomatal conductance. Consequently, growth responses of C_3 plants, including mangroves, could be enhanced under elevated atmospheric [CO_2] if growth in saline soils were limited by water, carbon or nitrogen. However, in a review of plant responses to elevated [CO_2] under saline conditions, Ball and Munns (1992) concluded that elevated [CO_2] would have little or no effect on growth if the soil water salinity were too high for a species to absorb water through the roots, if the carbon sink for growth was already saturated, or if saline conditions were to induce deficiencies in nutrients such as potassium. Indeed, growth of two mangrove species (*Rhizophora apiculata* and *R. stylosa*) in a multifactorial experiment, showed that WUE was enhanced by elevated [CO_2] under different combinations of salinity (125 and 350 mM NaCl) and humidity (43% and 86% relative humidity at 30°C); however, enhanced WUE did not necessarily translate into enhanced growth. Elevated [CO_2] enhanced growth when it was limited by low humidity at the leaves, but not when it was limited by high salinity at the roots (Ball et al., 1997). Similarly, Reef et al. (2015) showed that elevated [CO_2] did not alleviate limitations to growth of *Avicennia germinans* imposed by supra-optimal salinities, despite improvements in WUE. The results of these studies underscore complexities in the carbon, salt and water relations of species that are not captured in a single measure of WUE. Nevertheless, the results also offer insights into ways that future atmospheric [CO_2] could induce change in mangrove forest ecosystems along complex gradients in salinity and aridity.

7. Ecosystem-level water use efficiency of mangrove forests is comparably high as the leaf level

New observation techniques such as eddy covariance, sap flow sensors, dendrometer bands, and stable isotopes aid continuous monitoring of fluxes of energy, carbon dioxide and water vapor in mangrove ecosystems, thereby facilitating evaluation of water-carbon relations at an ecosystem level (Beer et al., 2009). According to the definition, ecosystem-level water use efficiency (WUE_e) is estimated by the ratio of gross primary production (*GPP*) to canopy transpiration (*T*) as shown in Table 1. However, availabilities of canopy transpiration data are limited for many ecosystems, particularly mangroves (Helbig et al., 2020). Instead, WUE can be estimated by alternative means using easily gaugeable variables, such as *GPP/ET* and *NEP/ET* where *NEP* is net ecosystem production. These terms were successfully applied to many upland ecosystems with two assumptions: first, evapotranspiration (*ET*) was mainly from T, and second, *NEE* was driven mainly by GPP (Beer et al., 2009; Hatfield & Dold, 2019). In contrast, those assumptions are not applicable in mangrove ecosystems where values of T/ET (Krauss et al., 2015; Liang et al., 2019) and NEP/GPP (Barr et al., 2010) are far less than one. Comparisons of different WUE_e calculations shows that GPP/T is significantly higher than the other estimations (GPP/ET, NEP/ET, and NEP/T), with the monthly-average GPP/T values ranging from 2.5 to 5 gC/kg H_2O in a subtropical mangrove forest (Fig. 3). These values of GPP/T are comparable to evergreen forests (Beer et al., 2009) but are almost double those of the commonly used term, GPP/ET. Use of the latter gives the misleading impression that mangrove forests are less water-use efficient than upland forest ecosystems (Xiao et al., 2013).

Calculations of GPP/T gave values consistent with independent measures of many parameters, such as ^{13}C (Farquhar, Ball, et al., 1982; Martin et al., 2010; Sternberg, 1992a), leaf stomatal traits (Liang et al., 2018) and leaf-level gas exchange (Naidoo & Willert, 1995; Sternberg, 1992), all of which reveal that mangroves generally have higher water use efficiency than neighboring upland trees.

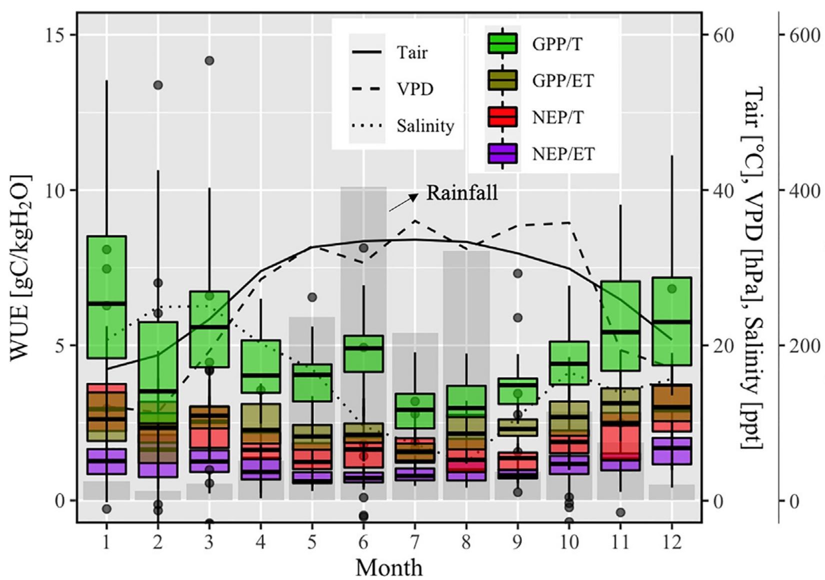

Fig. 3 Water-use efficiency of a mangrove ecosystem at Guangdong, China, 2012. The dominant mangrove species were *Aegiceras corniculatum*, *Bruguiera gymnorrhiza*, and *Avicennia marina*. The box plots were drawn from statistics of daily data in Cui et al. (2018) and Liang et al. (2019) where canopy transpiration values were modeled. The gray bars are rainfall. Salinity was measured in soil pore water, and rainfall for each month is the sum of daily values.

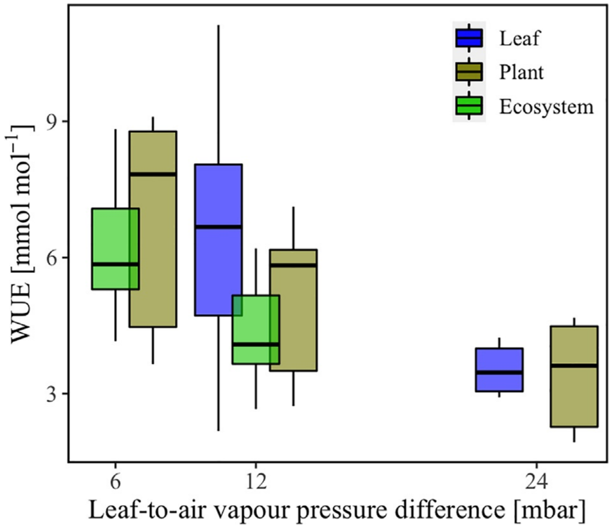

Fig. 4 Comparison of long-term (see Table 1) leaf-, plant-, and ecosystem-level water-use efficiency (WUE) under different humidity. The leaf-level WUE was calculated using data from Ball and Farquhar (1984b), and plant-level using the same data as in Fig. 2B, and ecosystem-level using data from the same source as in Fig. 3, but constrained to an irradiance of 400 W m^2 < net radiation to match light levels across the three studies.

A striking result of the present review, is the finding of similar long-term WUE in mangroves across leaf, plant and ecosystem levels when comparisons were constrained for similar values of D (Fig. 4). A significantly higher value of leaf-level WUE (13.5 ± 1.2 mmol mol^{-1}) at D of 6 mbar (Ball and Farquhar, 1984b) was not included in Fig. 4 because of the high probability of measuring errors in leaf gas exchange measurements made under such high humidity. Further, WUE at all levels of organization declined with increasing D (Figs. 3 and 4). These patterns, admittedly compiled from a limited number of mangrove species, indicate that water use efficiencies determined through stomatal behavior are conserved across scales from individual leaves to plants, and ultimately to ecosystems.

8. Unresolved complications and future directions

There are at least three major complications that require research to further understanding of the integration of water-use efficiency across levels of individual leaves, plants and mangrove forest systems.

First, calculations of stomatal conductance require accurate estimations of vapor pressure differences between the intercellular gas spaces within a leaf and the atmosphere. While atmospheric vapor pressure can be measured, the corresponding values within leaves typically are estimated by assuming that the vapor pressure in intercellular gas spaces is equivalent to the saturation vapor pressure above pure water at leaf temperature. However, the equilibrium saturation vapor pressure declines with decreasing water potential. At the low leaf water potentials at which mangroves generally function, under-saturation of the intercellular gas spaces would lead to an underestimation of stomatal conductance for a given transpiration rate and overestimation of transpiration rates for a given stomatal conductance (Canny, Wong, Huang, & Miller, 2011; Cernusak et al., 2018; Vesala et al., 2017).

Second, foliar absorption of atmospheric water could enhance water-use efficiencies over time periods significant to carbon gain and growth. For example, Nguyen et al. (2017) showed that upper limits to leaf rehydration and turgor generation in *Avicennia marina* declined with increasing soil water salinity if roots were the sole source of water supply to shoots, suggesting increasing requirements to access atmospheric water. Indeed, a capacity for foliar water uptake has been reported in a variety of mangrove species (Bryant et al., 2021; Hayes et al., 2020) with enhancement of water status enabling turgor driven growth (Schreel, Van de Wal, Herve-Fernandez, Boeckx, & Steppe, 2019; Steppe et al., 2018), recovery of leaf hydraulic conductance (Fuenzalida et al., 2019), and maintenance of carbon gain together with protection against stem hydraulic failure (Coopman et al., 2021). Use of atmospheric water would reduce carbon costs associated with uptake of soil water by roots and its transport to leaves, potentially enhancing plant level water-use efficiencies over the long term. These effects need to be taken into account in formulation of carbon and water budgets of mangroves.

Third, integration of carbon gain and water use by forest systems requires further development of the knowledge needed to interpret complex combinations of natural isotopic signatures. The issues arise because no single isotopic signature provides all the information required to understand water use and water-use efficiencies, and because the isotopic composition of some substances, such as water, can vary because of both environmental conditions and responses of plants to them. Deciphering this complex isotopic code is fundamental for analyses of mangrove forest function under past and current conditions that are needed to develop process-based modelling tools for management of coastal mangrove resources under changing environmental conditions. For example, the oxygen isotopic composition of mangrove cellulose is related to leaf traits such as stomatal density and leaf succulence (Liang et al., 2018). Combining oxygen with carbon isotopic compositions will provide more comprehensive information of mangrove water-use efficiency and water use in response to environmental conditions.

9. Conclusion

In this review, we have clarified definitions of long-term, instantaneous, and intrinsic water-use efficiency and distinguished estimations of WUE at levels of individual leaves, plants and ecosystems. Using this framework, we showed consistency between the high WUE typically measured in mangrove leaves and those estimated over time scales significant for the growth of plants and the functioning of ecosystems. Finally, we drew attention to a few unresolved complications that affect estimations of WUE, notably effects of unsaturated intercellular vapor concentrations on estimations of conductance and transpiration, effects of foliar water uptake on long-term water-use efficiency of plants, and the need to develop methods to further exploit the information rich signatures of stable isotopes for better understanding of water use characteristics of mangrove systems under past, present and future environmental conditions.

Acknowledgments

The authors acknowledge the Ngunnawal people, the traditional custodians of the land on which this work was conducted. We pay our respects to them and their cultures and to elders both past and present. J.L. and M.C.B. thank the Australian Research Council for support through Discovery Project Grant DP180102969, and G.D.F. thanks the ARC for support through DP210103186.

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